WEEK 4 – ASSIGNMENT

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Title:

Azure Cloud Fundamentals and Data Pipeline Implementation using ADF

Objective:

To understand Azure cloud concepts and build a complete data pipeline using Storage Account and Azure Data Factory.

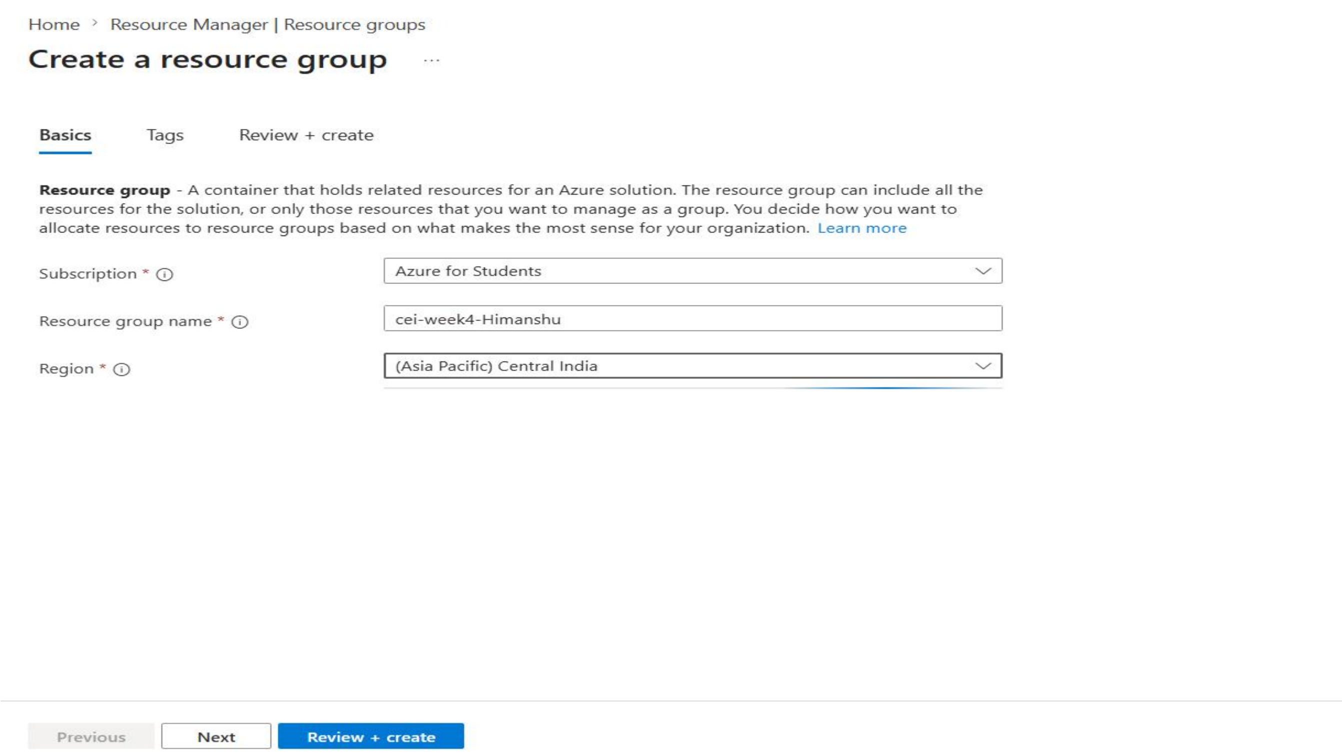
Assignment Tasks

Task 1:

- Explore Azure Portal
- Create a Resource Group

Deliverable:

- Screenshot of Resource Group



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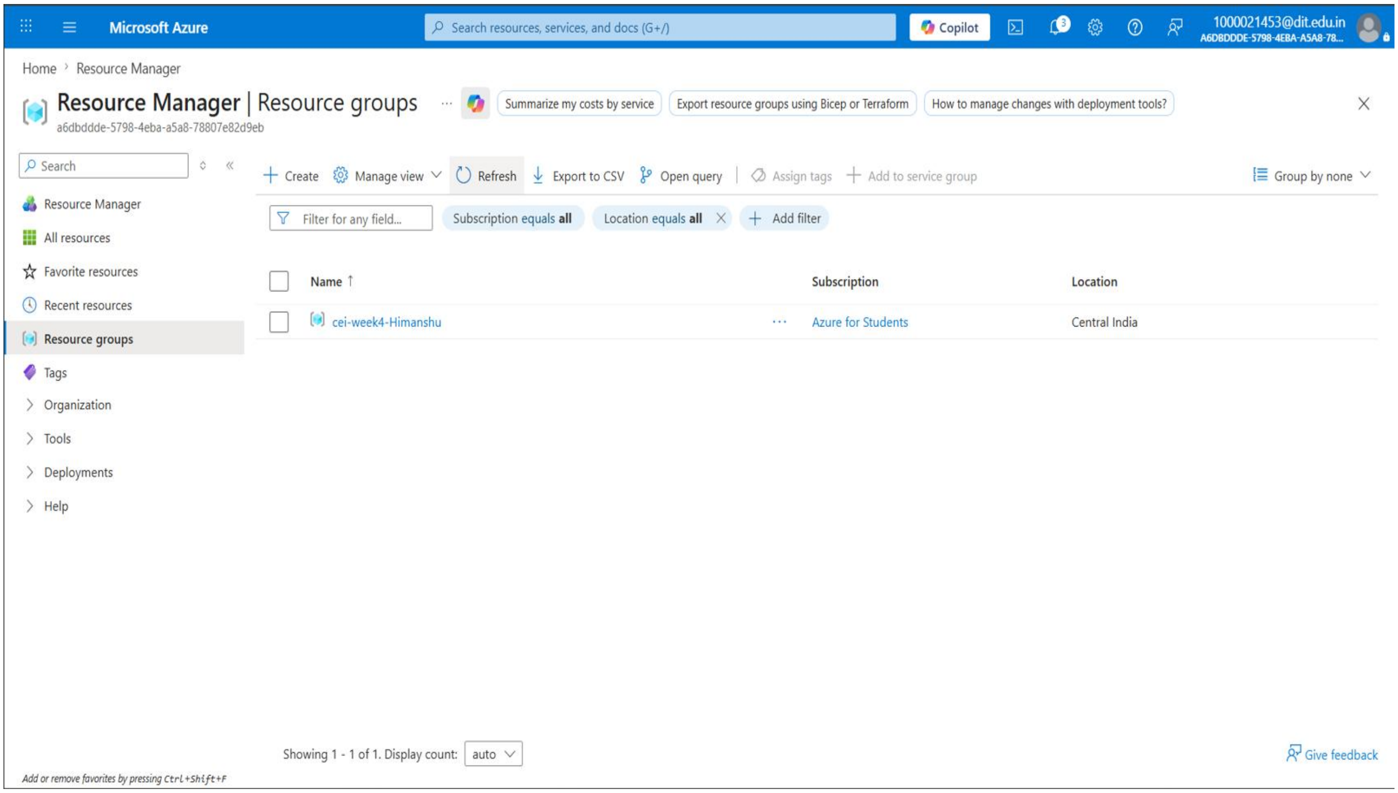

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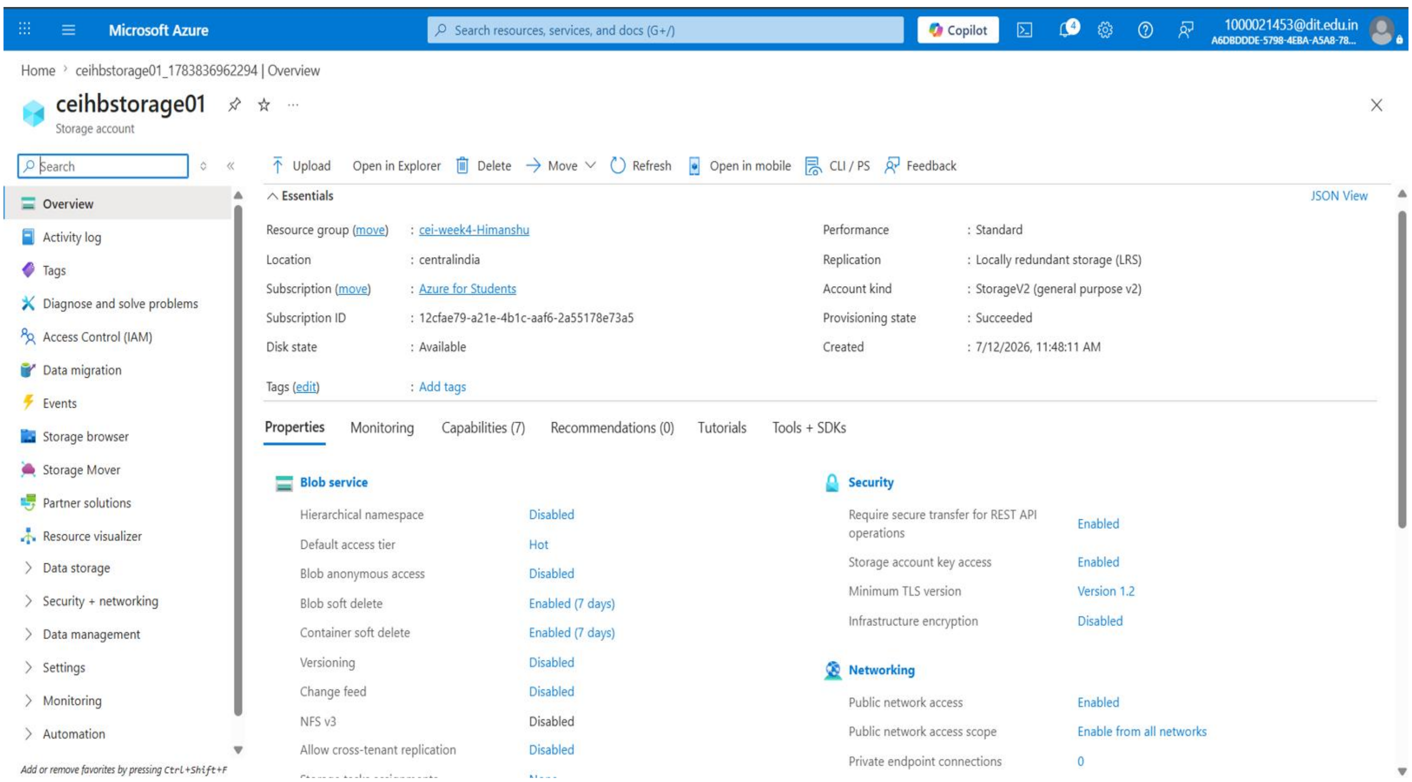
Task 2: Storage Setup

Do:

- Create a Storage Account
- Create a Blob Container
- Upload a CSV file

Deliverable:

- Screenshot of container with uploaded file



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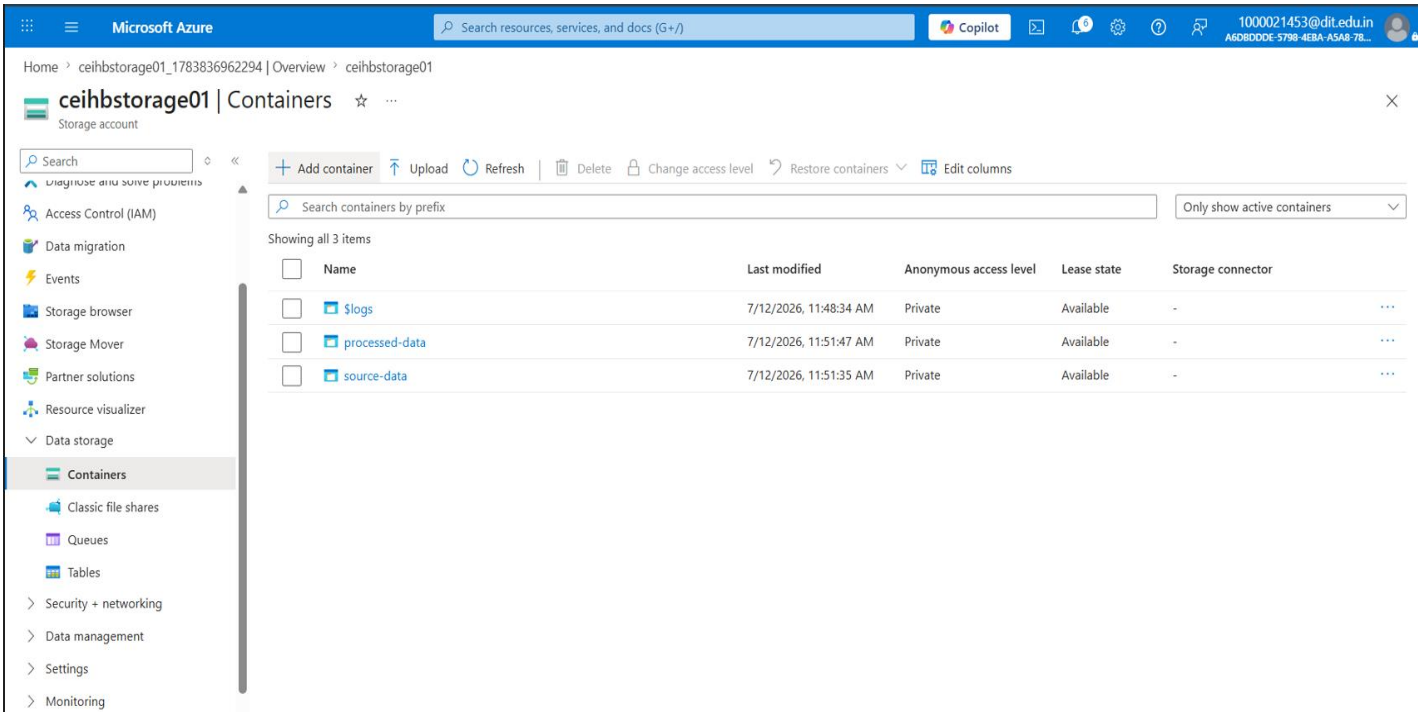

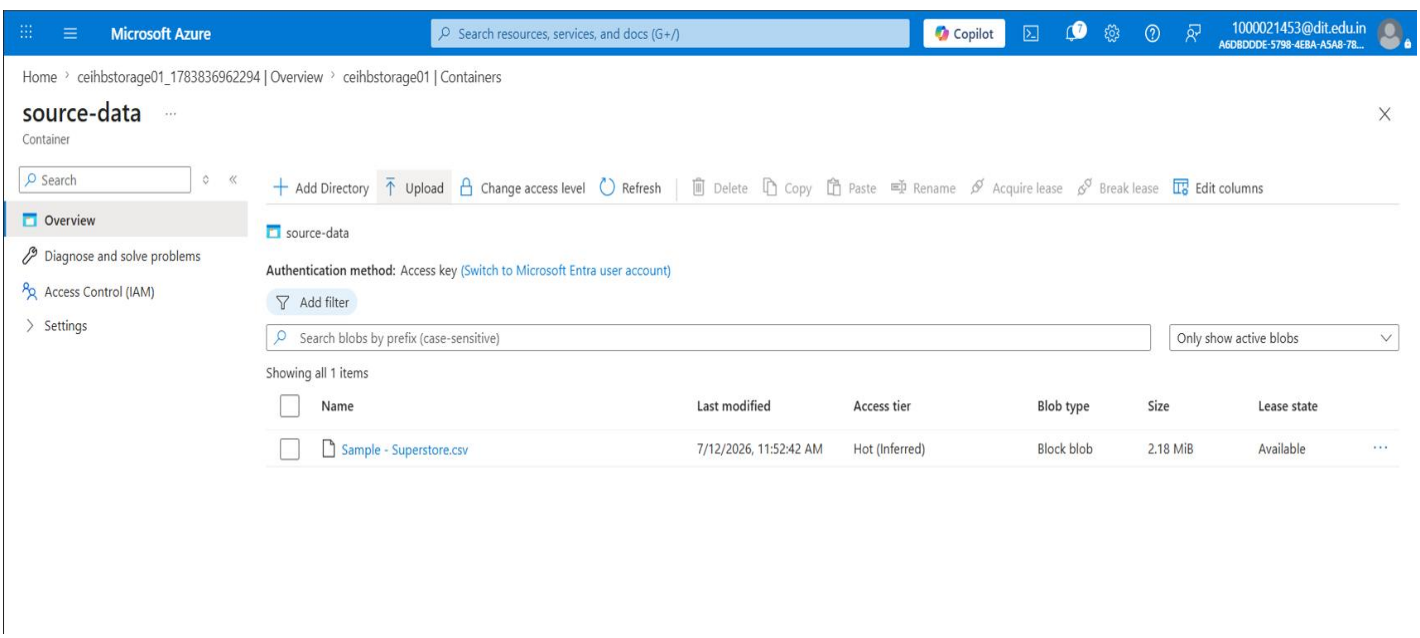

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Task 3: ADF Basics

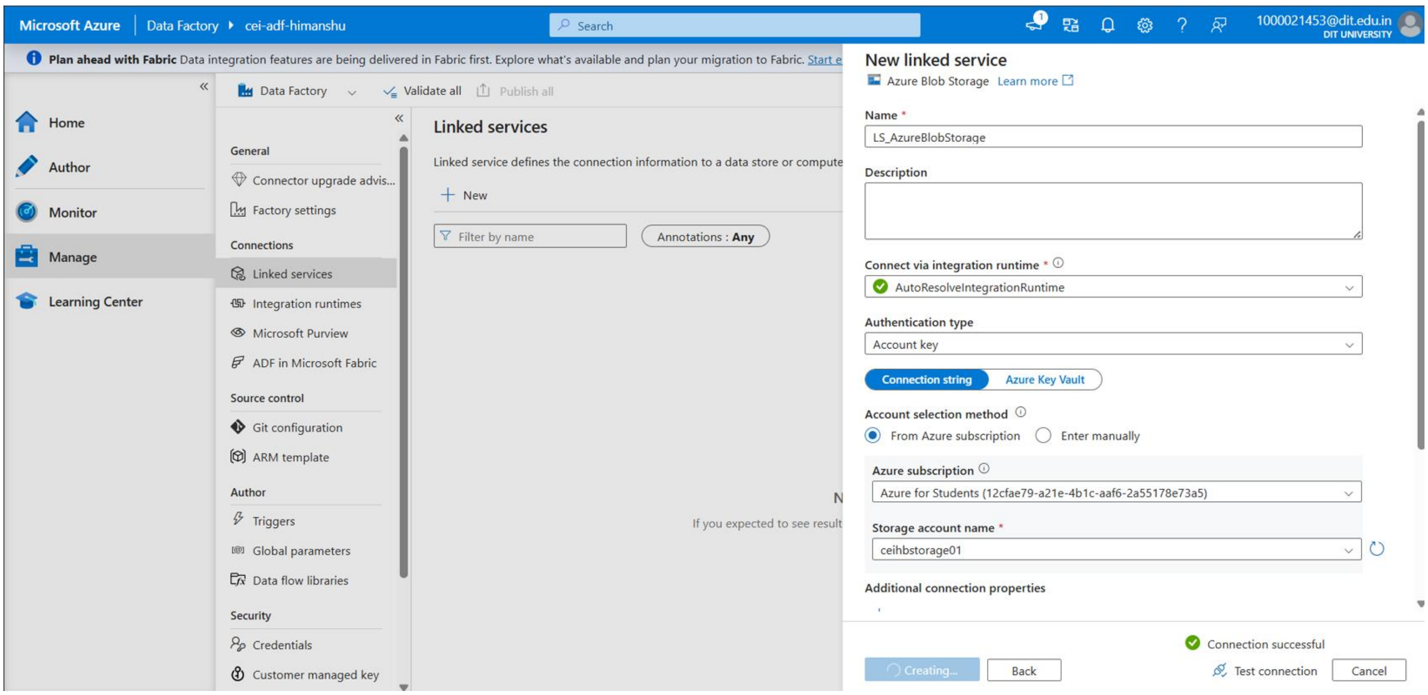
Do:

- Create Azure Data Factory
- Explore ADF UI (Author, Monitor, Manage)
- Create Linked Service (Blob Storage)
- Create datasets (source + destination)
- Use Get Metadata activity

Deliverable:

- Screenshot of Linked Service
- Screenshot of dataset
- Screenshot of Get Metadata activity

a.) Screenshot of Linked Service



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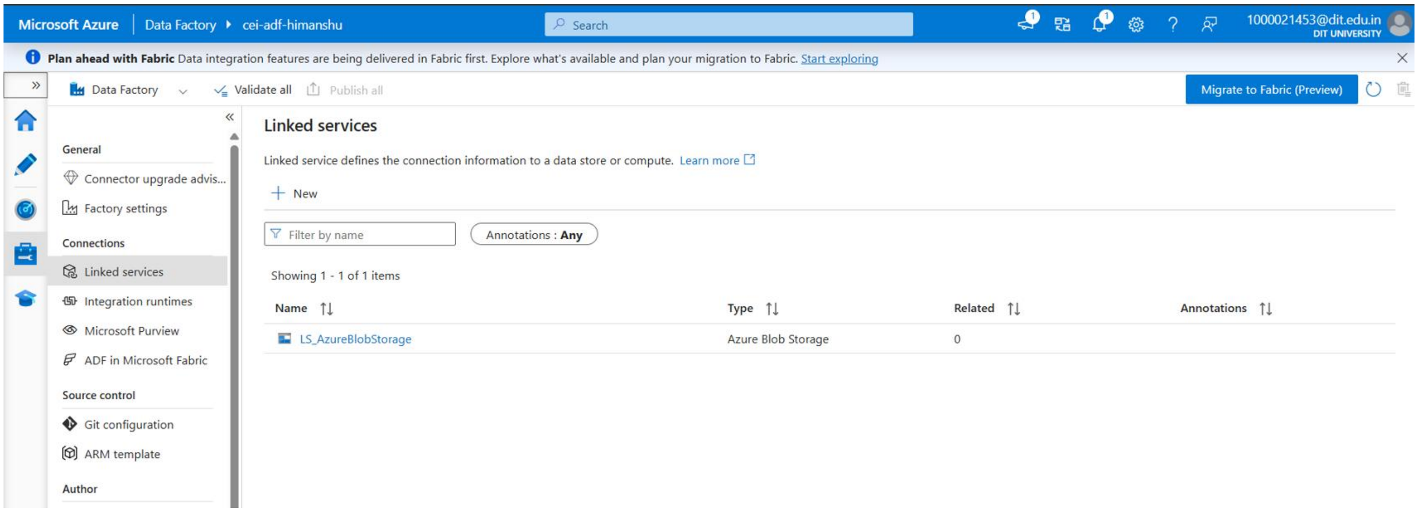
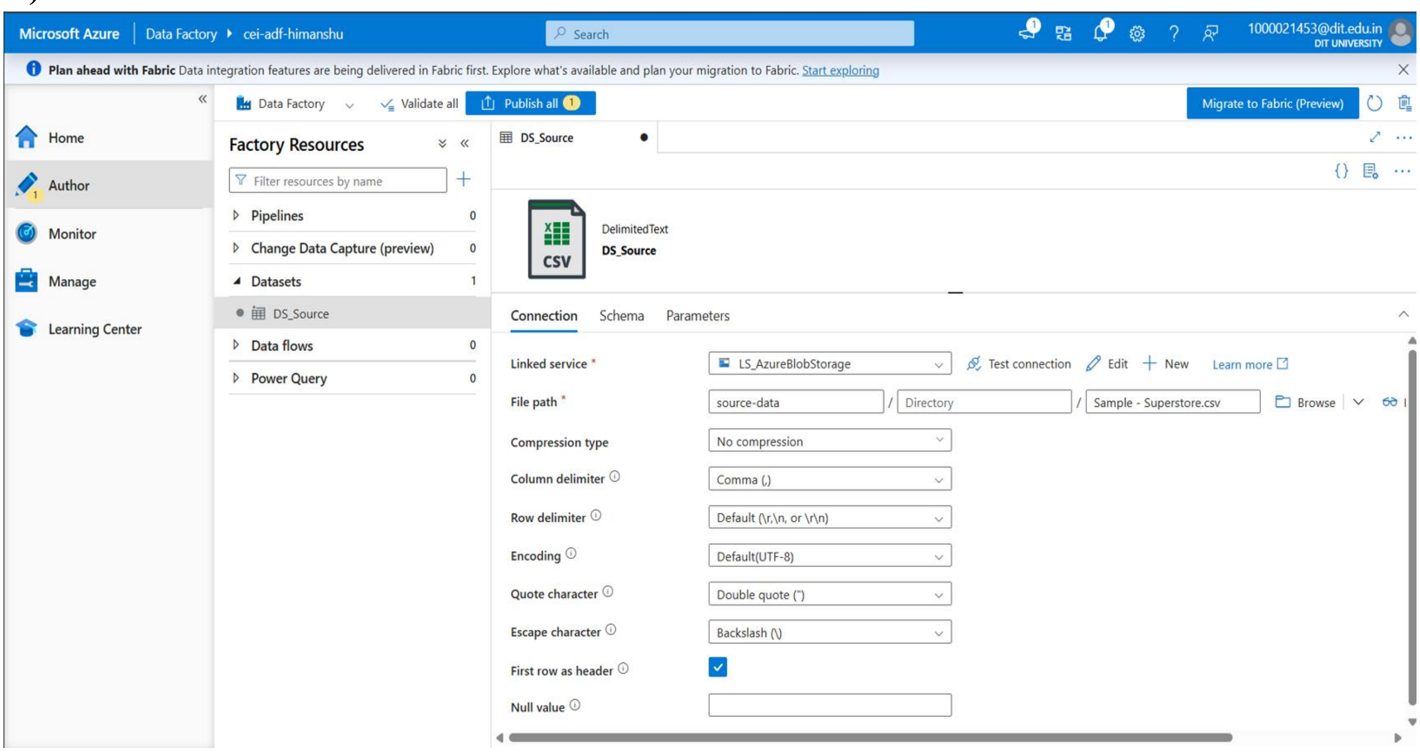


b.) Screenshot of DataSet



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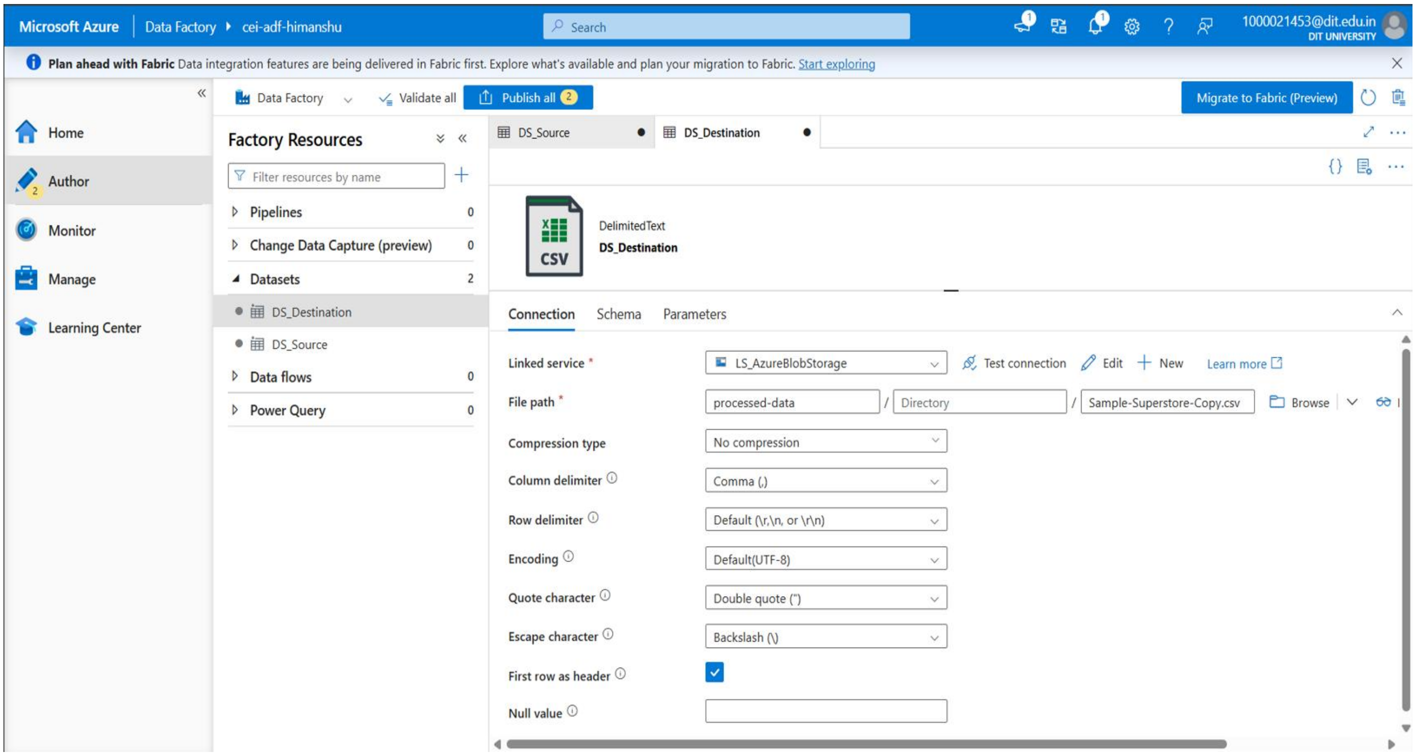
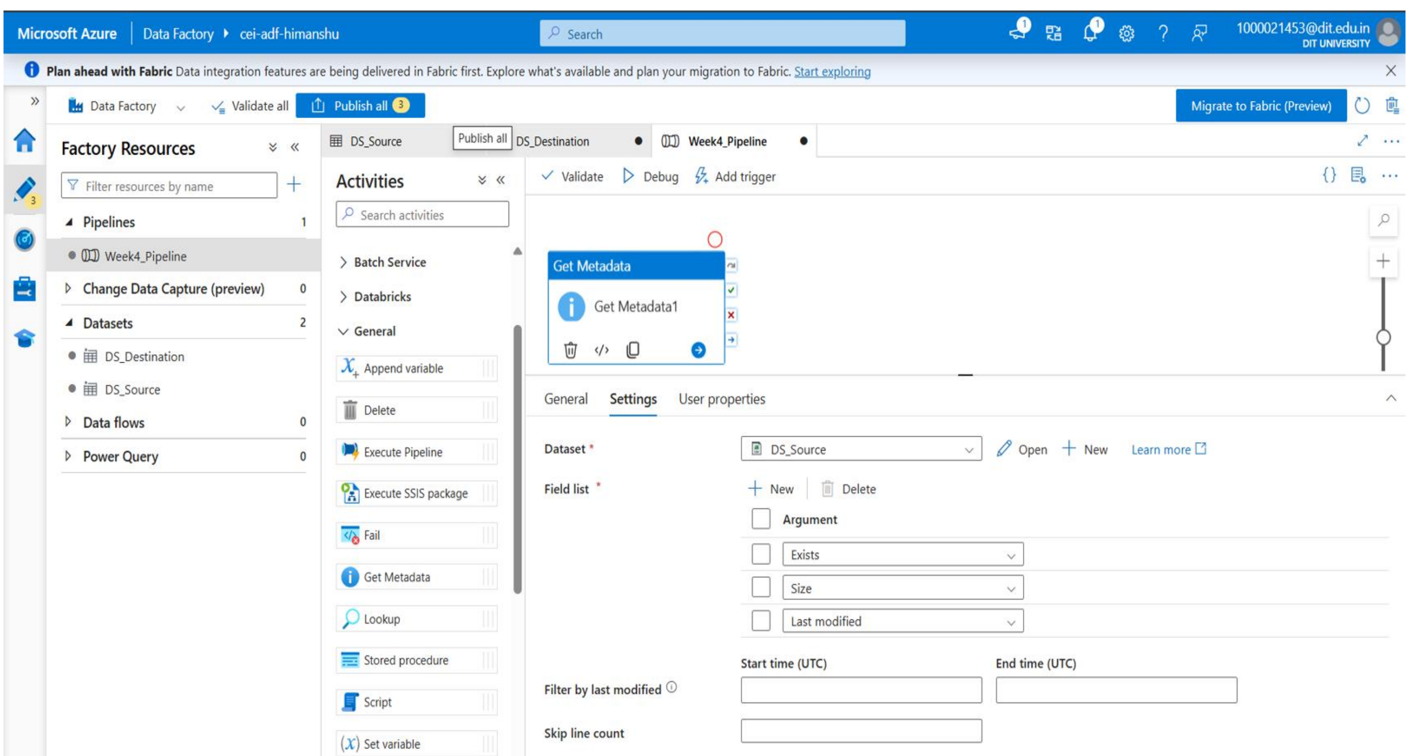


c.) Screenshot of Get Metadata Activity



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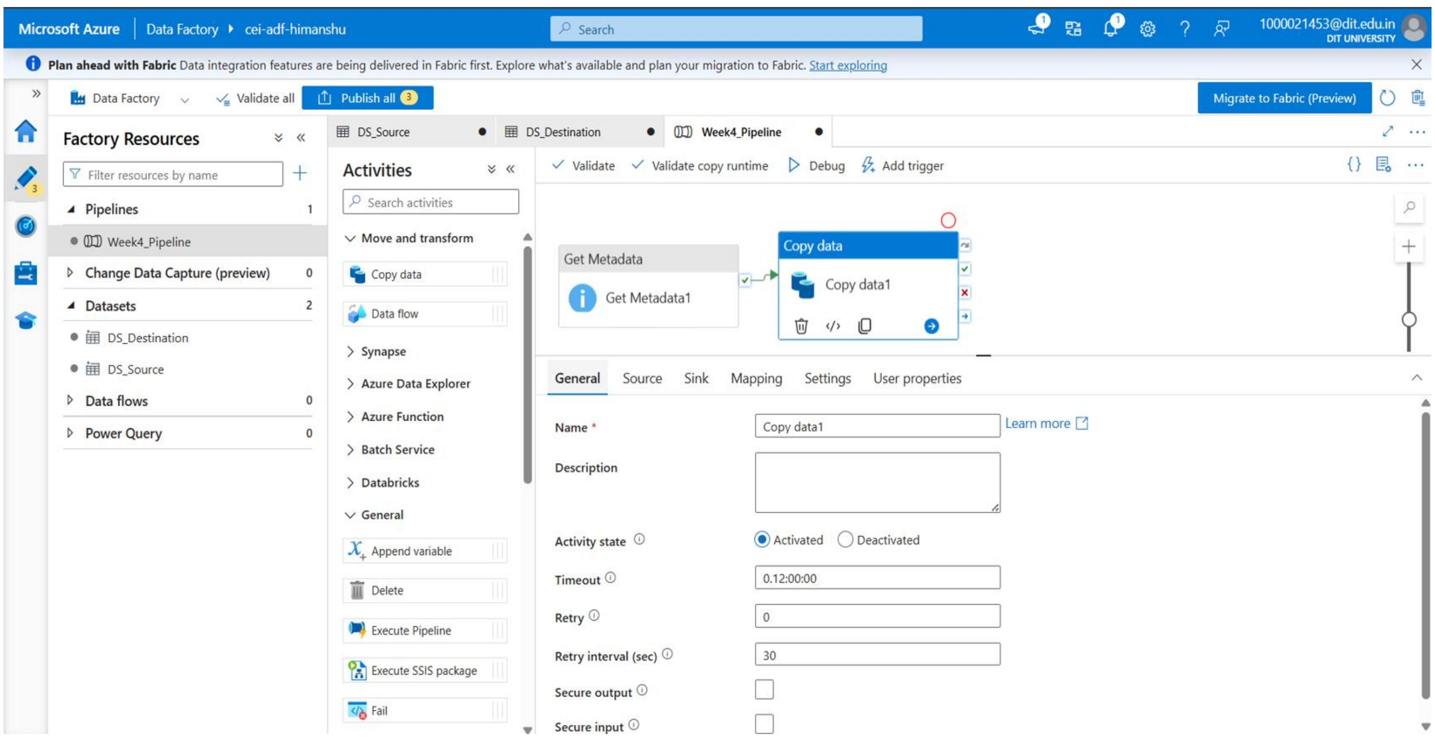
Task 4: Pipeline Development

Do:

- Create pipeline using Copy Data activity
- Configure source and destination
- Add ForEach activity (optional if multiple files)

Deliverable:

- Screenshot of pipeline design



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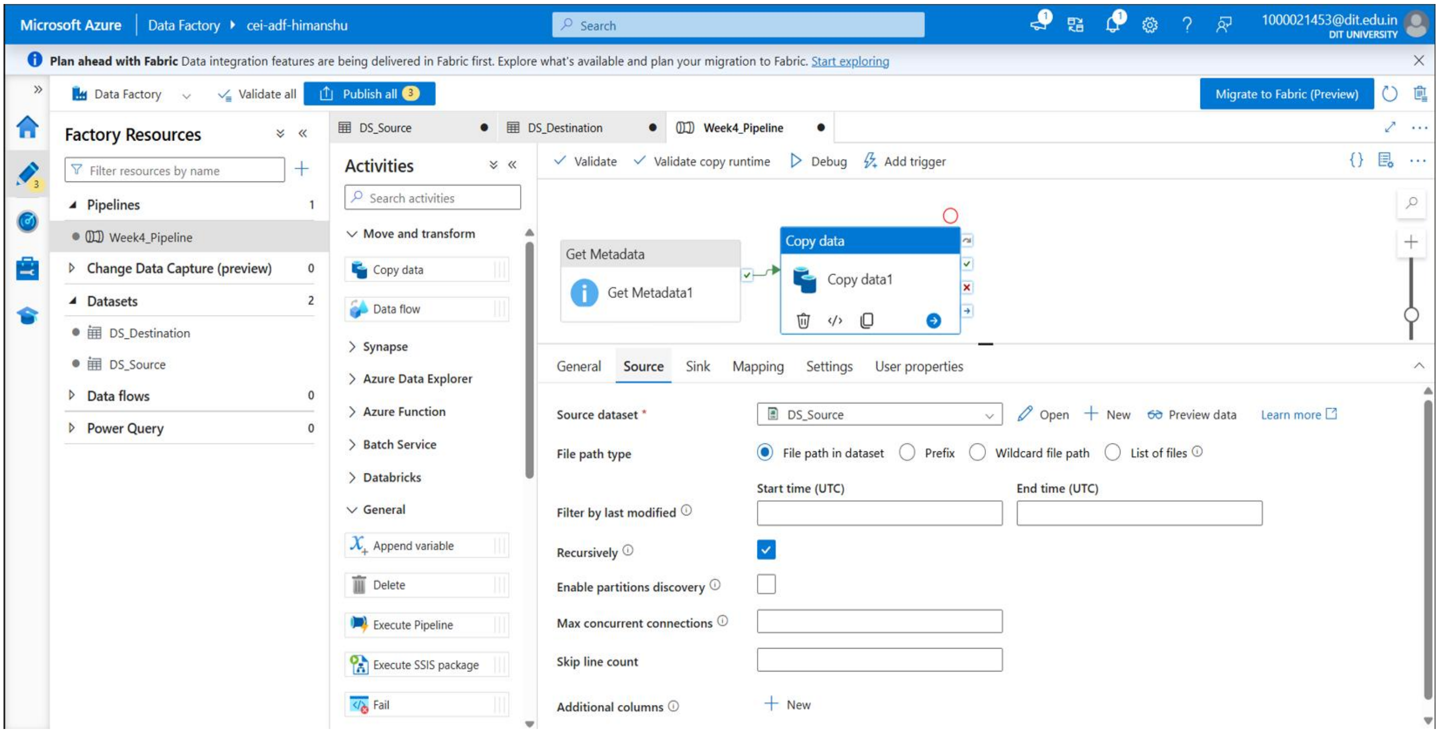

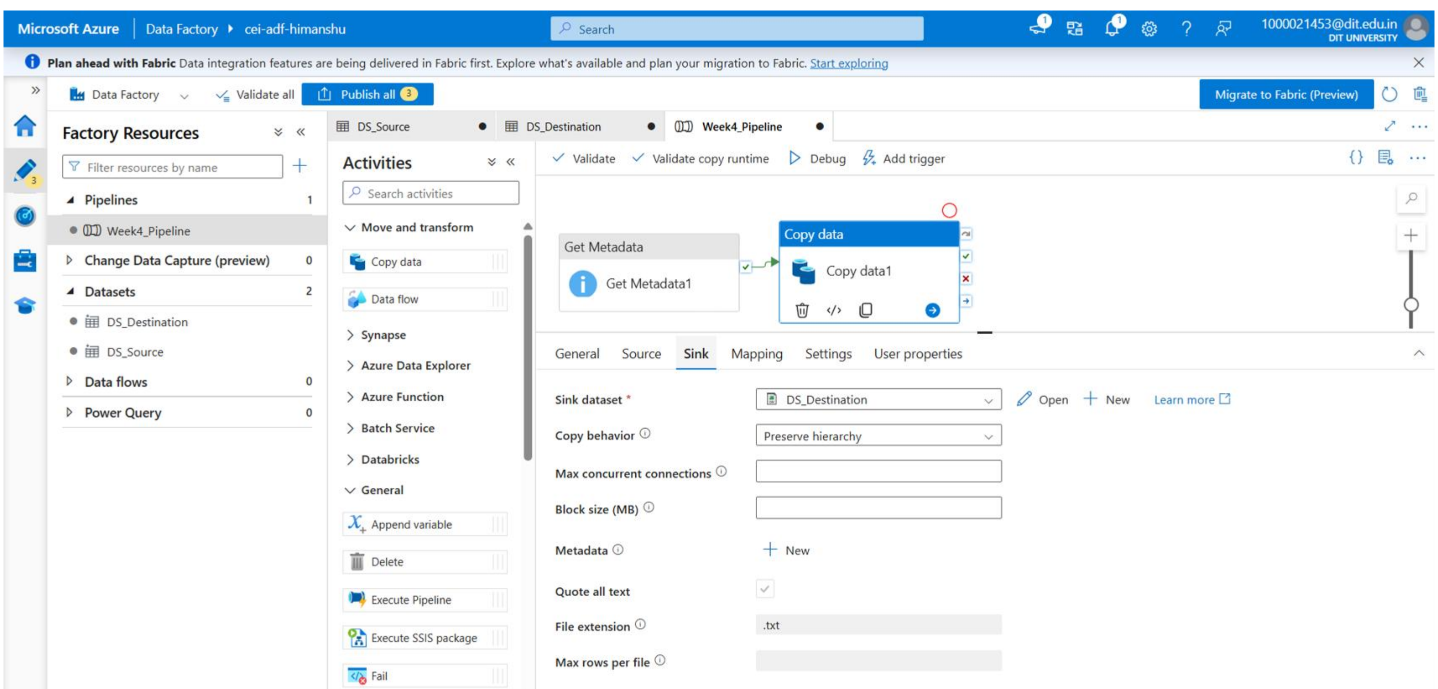

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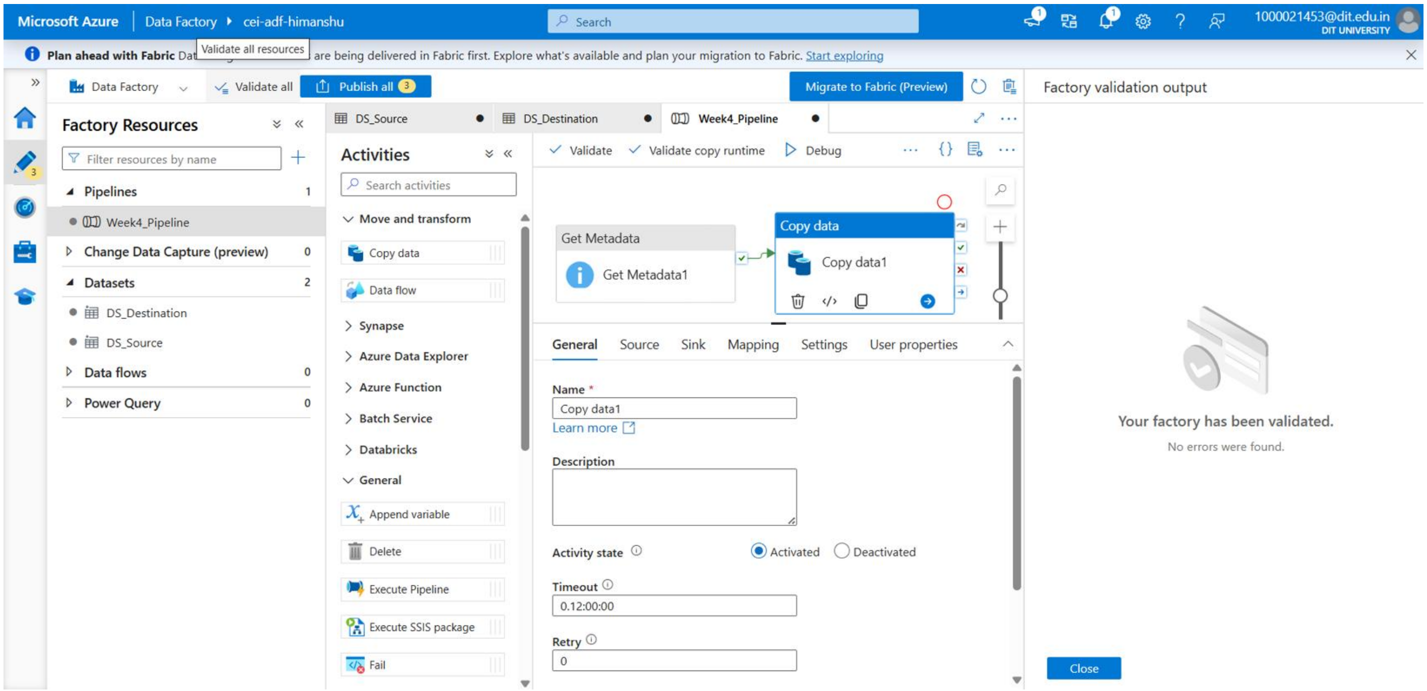
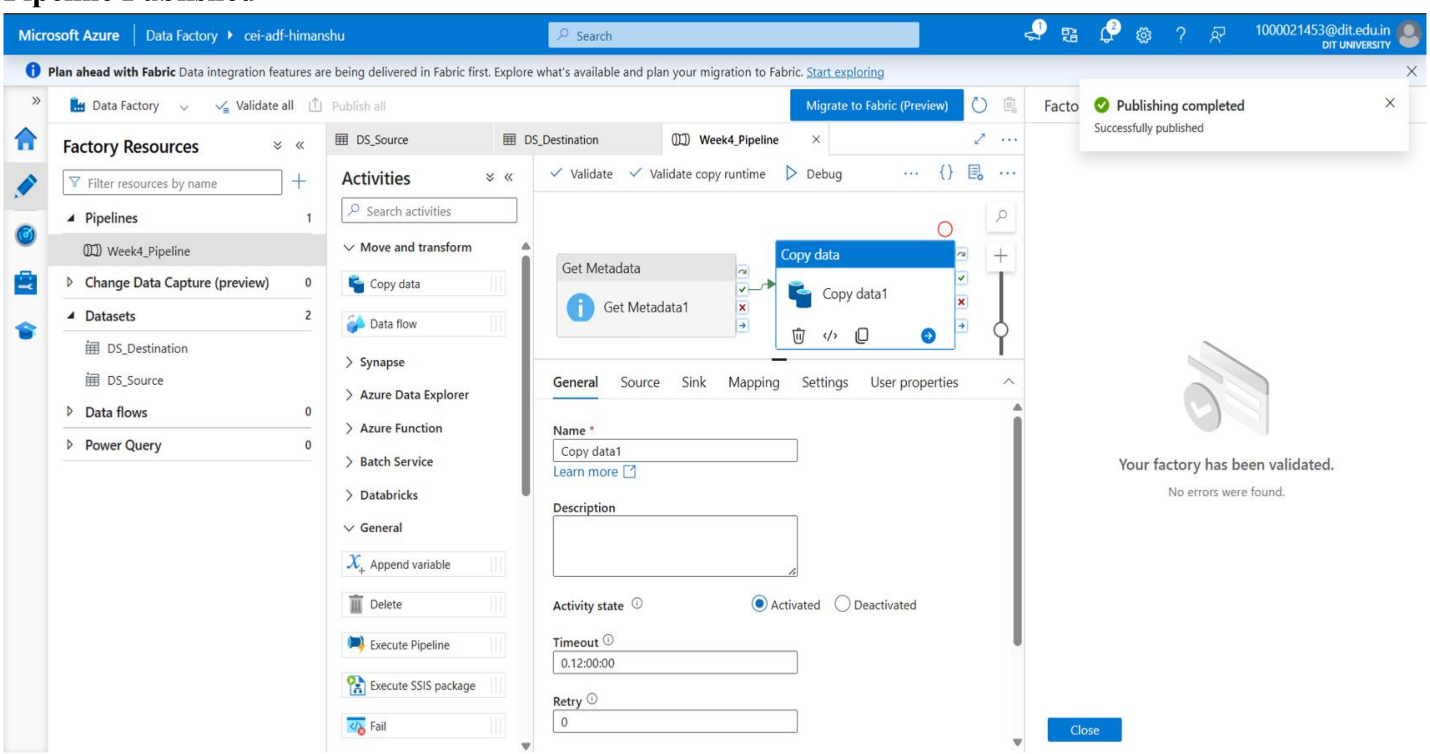
Pipeline Validation



Pipeline Published



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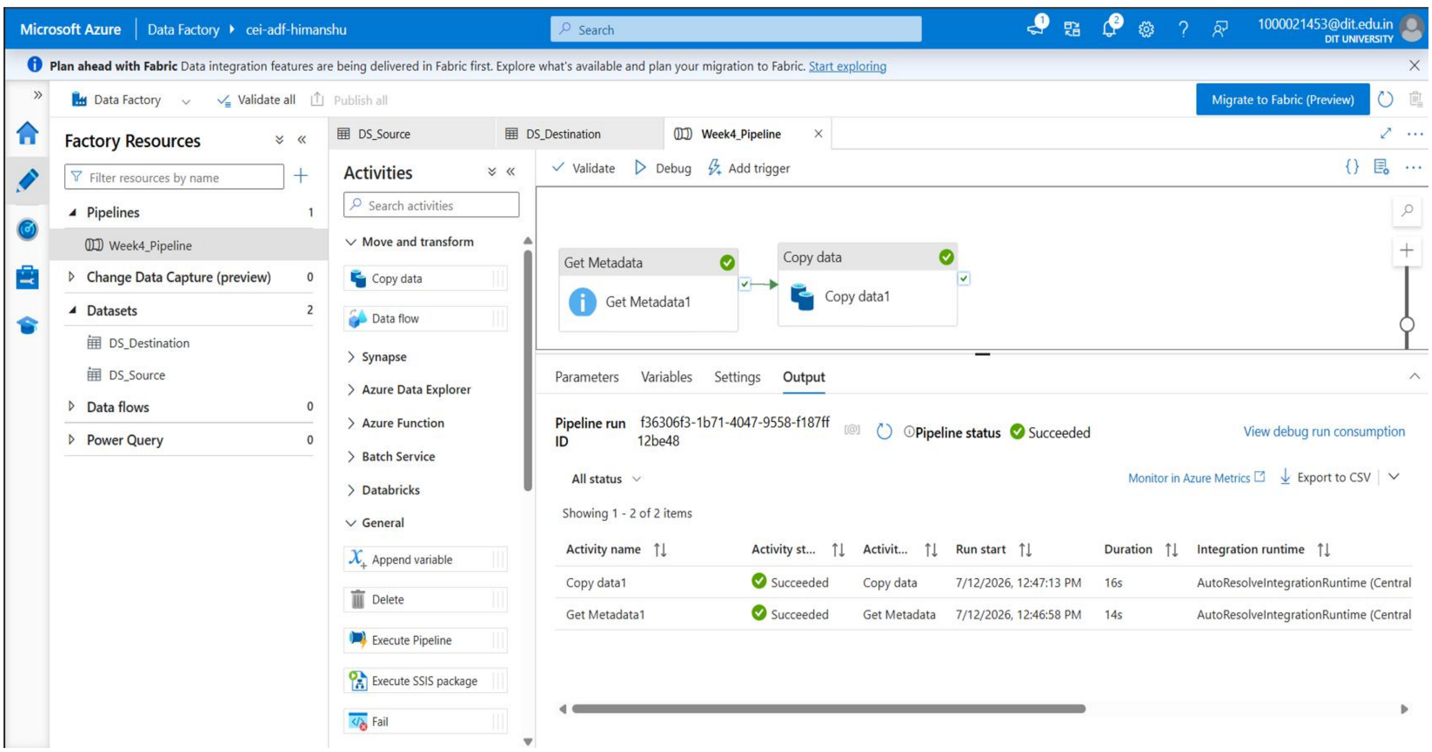
Task 5: Pipeline Execution

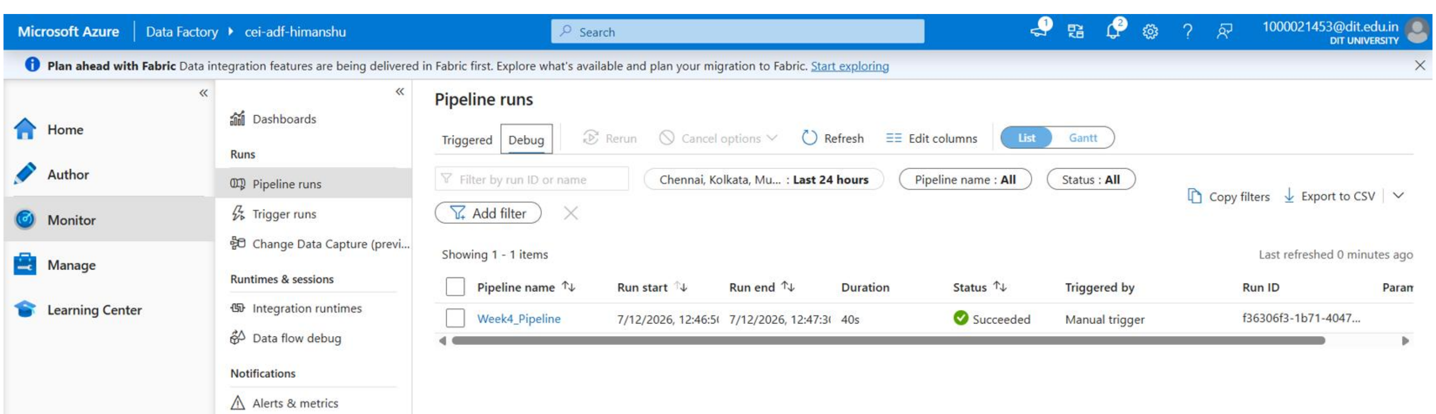
Do:

- Run pipeline using Debug/Trigger

Deliverable:

- Screenshot showing pipeline execution (Succeeded)





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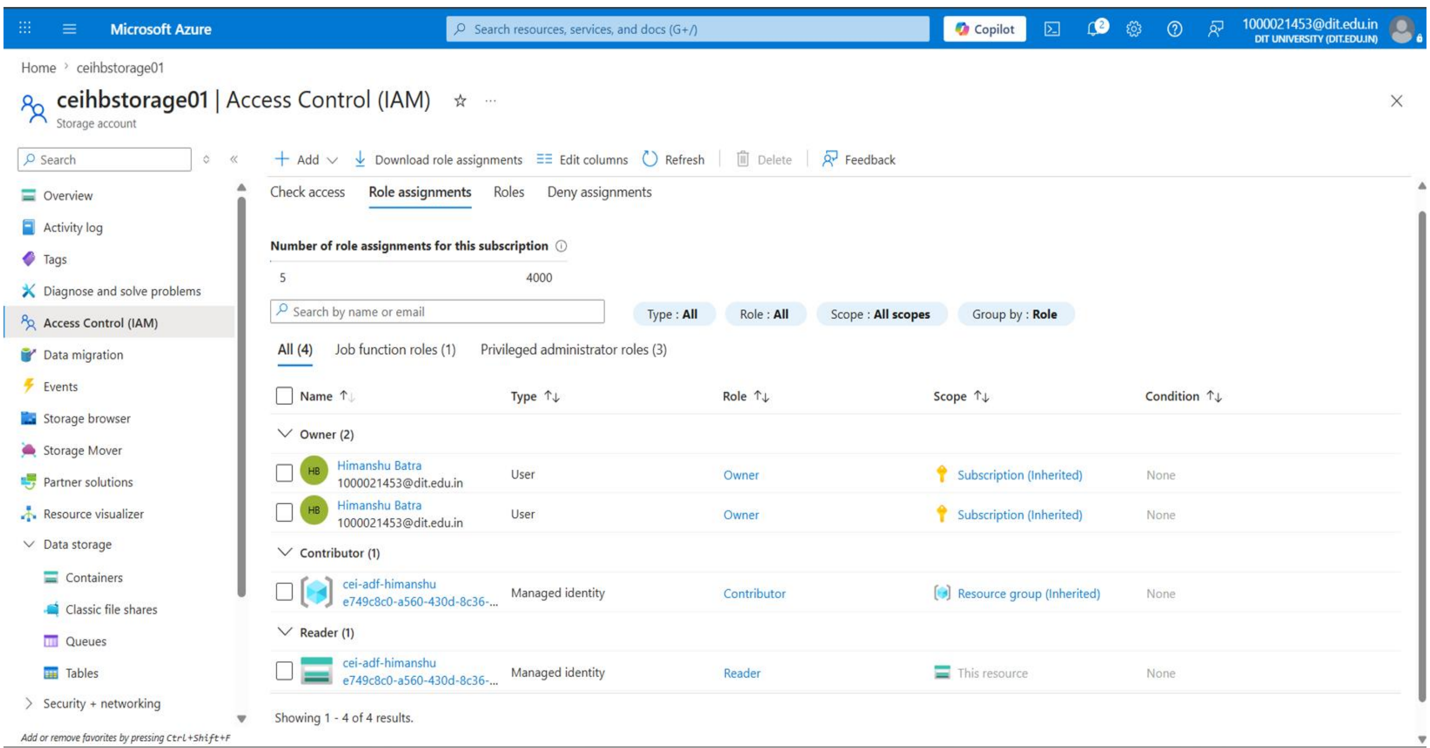
Task 6: IAM Roles

Do:

- Assign roles:
  - Reader
  - Contributor
- Provide access to ADF for Storage

Deliverable:

- Screenshot of role assignment



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Pipeline Execution Results

The Azure Data Factory pipeline was successfully validated, published, and executed using the **Debug** option. The **Get Metadata** activity completed successfully by validating the source CSV file's metadata, including its existence, size, and last modified timestamp. Following successful validation, the **Copy Data** activity transferred the CSV file from the `source-data` Blob container to the `processed-data` Blob container without any errors.

Execution Status

- **Pipeline Status:** ✅ Succeeded
- **Get Metadata Activity:** ✅ Succeeded
- **Copy Data Activity:** ✅ Succeeded
- **Metadata Validation:** ✅ Successful
- **Data Copy:** ✅ Successful
- **Output File:** Successfully created in the `processed-data` Blob container.

Observations

- The pipeline executed without any runtime errors.
- Source file metadata was successfully retrieved before data transfer.
- The destination file was generated successfully in Azure Blob Storage.
- End-to-end data movement between source and destination containers was completed successfully.

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Mini Project (Final Task)

Objective

Built an end-to-end Azure Data Factory pipeline to copy a CSV file from Azure Blob Storage while validating metadata before processing.

Implementation

- Created an Azure Storage Account and Blob Containers.
- Uploaded the Superstore CSV file to the source container.
- Configured Azure Blob Storage Linked Service.
- Created Source and Destination Datasets.
- Implemented a Get Metadata activity to validate the source file.
- Used Copy Data activity to copy the CSV into the destination container.
- Assigned Reader and Contributor IAM roles to the Azure Data Factory Managed Identity.
- Successfully executed and monitored the pipeline.

Result

- ✅ Source CSV successfully detected.
- ✅ Metadata validation completed successfully.
- ✅ CSV copied to the destination container.
- ✅ Pipeline executed without any errors.
- ✅ End-to-end Azure Data Factory pipeline completed successfully.

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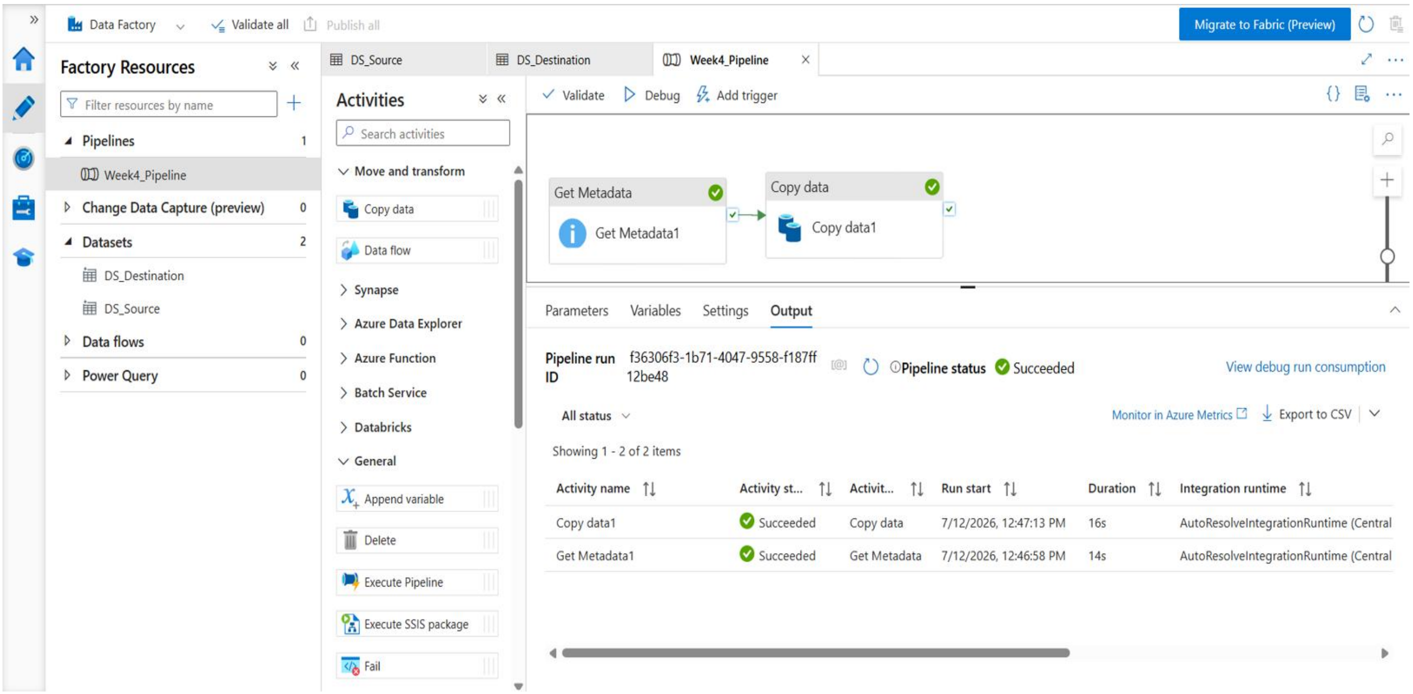
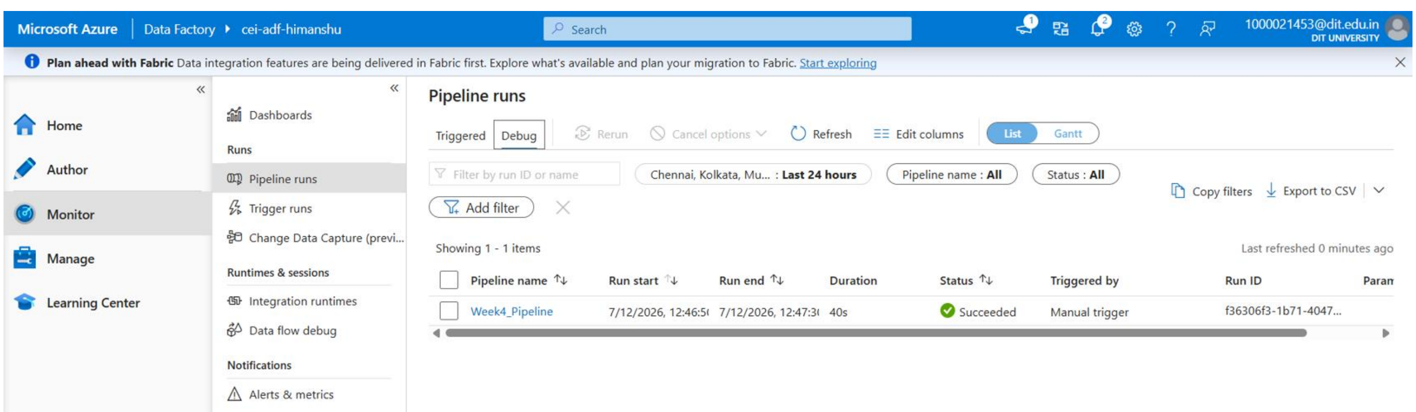
a.) Screenshot of Pipeline Successfully Executed





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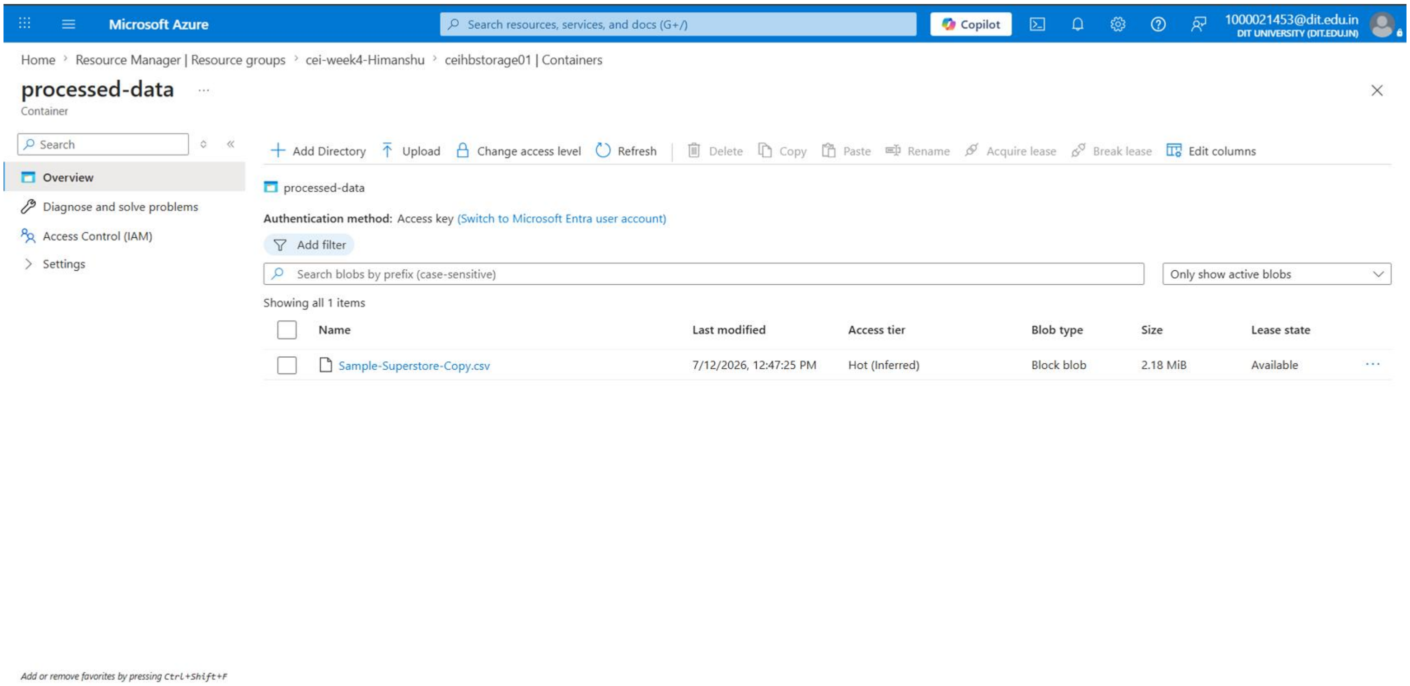
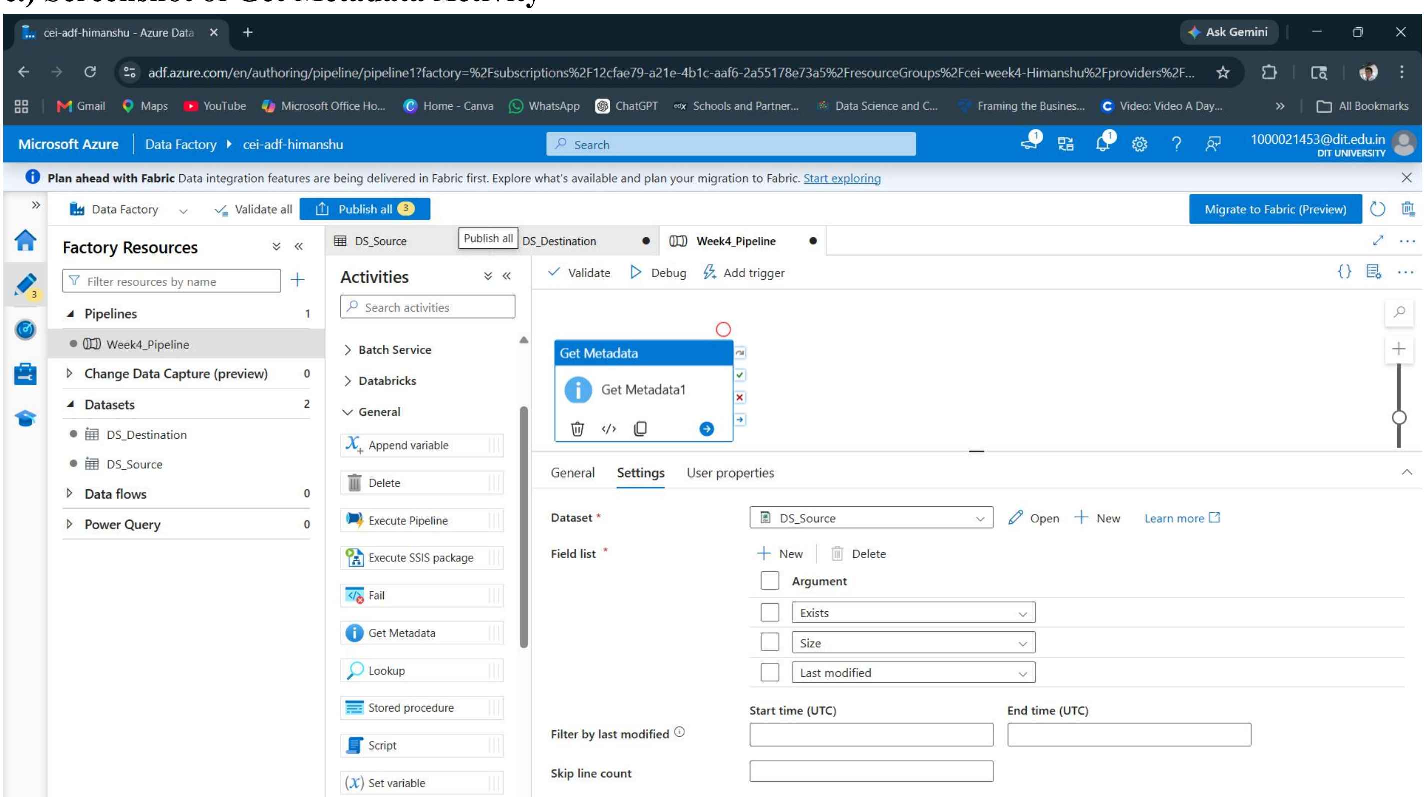
b.) Screenshot of Data Successfully Copied to Destination



c.) Screenshot of Get Metadata Activity



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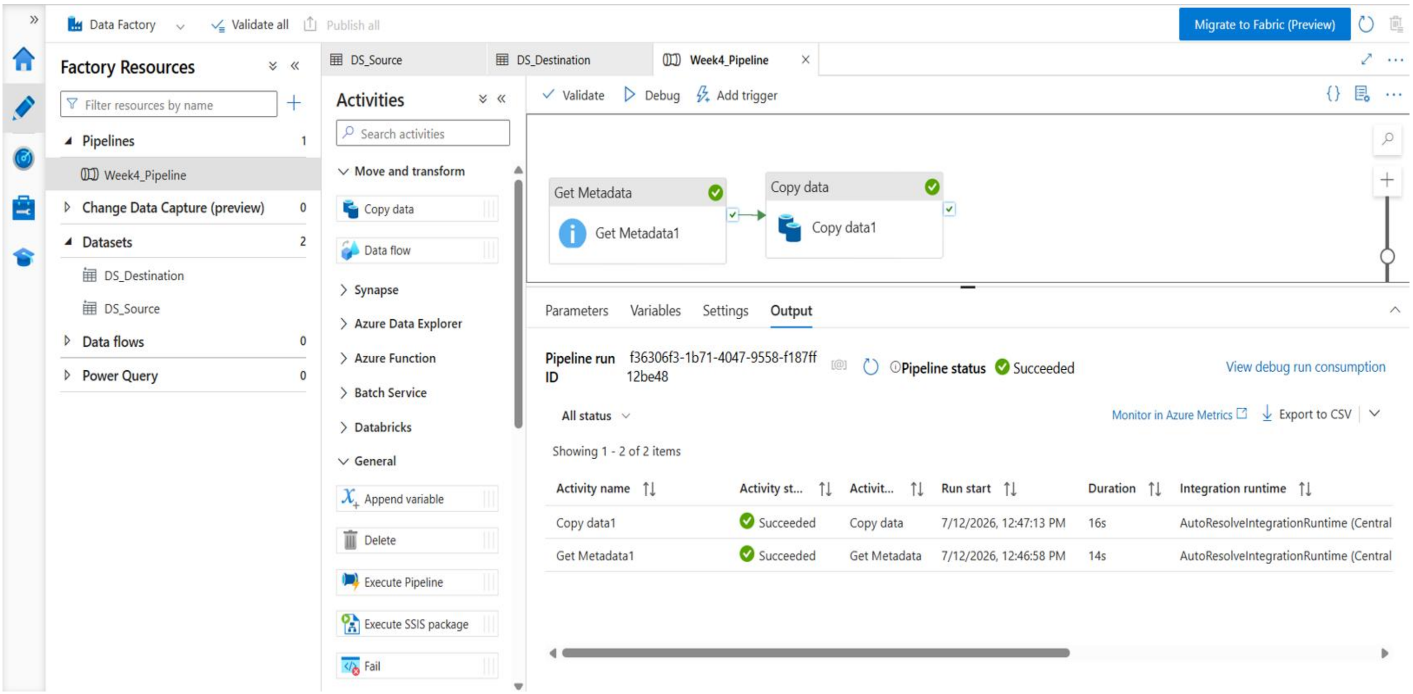
d.) Screenshot of Metadata Successfully Validated



Mini Project Summary

An end-to-end Azure Data Factory pipeline was developed to read a CSV file from Azure Blob Storage, validate its metadata using the **Get Metadata** activity, and copy it to a new destination container using the **Copy Data** activity. IAM roles (Reader and Contributor) were configured to enable secure access between Azure Data Factory and Azure Storage. The pipeline executed successfully, and the copied file was verified in the destination (`processed-data`) container, confirming successful data transfer and metadata validation.